Multi-Dataset Conditional Text Generation for Emotion Augmentation in Sentiment Classification

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Abstract. Sentiment and emotion classification systems can be limited by acute imbalance of classes, mixed taxonomy of labels and lack of representation of minority emotions. The paper proposes a single pipeline that will solve these issues through integrating emotion-conditioned text generation with transformer-based classification. Three benchmark datasets, GoEmotions, TweetEval, and DAIR Emotion, are used and underrepresented emotion-specific samples are synthesized with SmolLM-135M-Instruct using explicit emotion control tokens. Finally, RoBERTa classifiers are fine-tuned with the augmented data and hyperparameter optimization of Optuna. In all datasets, there is a steady increase in macro-F1 and minority-class recall with macro-F1 scores of 0.83 on GoEmotions, 0.79 on TweetEval, and 0.97 on DAIR Emotion. These findings suggest that when the emotion is conditioned to augmentation, generalization is enhanced in the classification of multiple datasets of emotions, especially in the sparse emotion categories.

Keywords: Emotion-conditioned augmentation · Sentiment analysis · Conditional text generation · Transformers · RoBERTa · Multi-label classification

1 Introduction

The application of NLP is fundamentally based on emotion and sentiment recognition, which operate on chatbots and social media analysers as discussed by Wolf et al. [12]. Though transformer-based models have become better, three current challenges exist in order to curtail performance. First, there are extreme form of the imbalance present in a class which biases the common emotions, i.e. joy and neutral against the minority emotions i.e. fear, surprise and love as well. Second, cross-domain learning and evaluation is problematic by heterogeneity of the label taxonomies across datasets in learning and evaluation of datasets with other domains problematic to cross-domain learning and evaluation. Third,

the emotions (fine-grained) are frequently not built on plentiful evidence, which cannot then be strongly modeled as a minority feeling.

Barbieri et al. [3], Demszky et al. [5], and Saravia et al. [9] present benchmark datasets characterized by a strong skew. Stronger gradients are given to dominant classes, and less frequent emotions are underrepresented hence the bias of the encoders and poorer generalization. Common classical mechanisms of augmentation like back-translation by Sennrich et al. [10], synonym replacement, and embedding-space oversampling introduced by Chawla et al. [4] address these problems only on a case-by-case basis and can cause artifacts which decrease fidelity of labels.

In order to overcome such shortcomings we introduce a multi-dataset emotion-conditioned augmentation model combining label-conditioned text generation and transformer-based classification. Minority classes are encouraged based on an instruction-conditioned generator with explicit emotion labels and classifiers are fine-tuned with Optuna as described by Akiba et al. [1]. The purpose of such strategy is to maintain the intent of affect and build a lexicon and contextual diversity on behalf of underrepresented label space. End-to-end reproducibility is maintained based on publicly published notebooks by Patrick [8]. In this paper, the general lettering will be abbreviated as emotion-conditioned augmentation and the label-controlled synthesis mechanism as emotion-conditioned text generation.

2 Related Work

2.1 Emotion Classification Datasets

Large annotated corpora have led to advances over emotion and sentence classification. The annotations of GoEmotions by Demszky et al. [5] also report the fine-grained annotation of more than 58,000 Reddit comments of 27 emotions. TweetEval by Barbieri et al. [3] is more concerned with sentiment classification of short, informal Twitter text, and DAIR Emotion by Saravia et al. [9] is more concerned with defining six basic emotions: sadness, joy, love, anger, fear, and surprise with intermediate balance. Although these datasets may have different granularities and domains, similar patterns of limitations can be observed: the most common ones are neutral and joy, but others, such as fear, surprise, grief, and other minor ones, are underrepresented and are sparse.

2.2 Text Data Augmentation

Classical augmentation techniques, including synonym replacement, random insertion or deletion, SMOTE-inspired embedding-space oversampling by Chawla et al. [4], and neural back-translation by Sennrich et al. [10], are widely used for text classification. Wei and Zou [11] proposed Easy Data Augmentation (EDA) and demonstrated that simple lexical perturbations such as random insertion, deletion, and swap can improve robustness in low-resource settings. However,

these methods may distort emotional context, induce semantic drift, or introduce artifacts that reduce fluency and label consistency. Emotion-conditioned augmentation provides a more targeted alternative by explicitly steering generated text toward desired affective labels. Anaby-Tavor et al. [2] show that neural text generation can produce useful task-specific augmentation, motivating the controlled generation strategy used in this work.

2.3 Transformer Models for Classification

Transformer architectures such as RoBERTa proposed by Liu et al. [7] achieve strong performance across NLP tasks as noted by Wolf et al. [12], but emotion and sentiment classification remain challenging due to dataset heterogeneity and imbalanced training signals found in Demszky et al. [5], Barbieri et al. [3], and Saravia et al. [9]. We therefore use Optuna-based hyperparameter search by Akiba et al. [1] to tune fine-tuning settings and improve macro-F1 and minority-class recall.

3 Methodology

The article uses a two-piping system to assess the effectiveness of emotion-conditioned data augmentation. The methodology is divided into four main steps: (1) Preprocessing and Imbalance control of three heterogeneous benchmark datasets; (2) Training a label-conditioned generative model (SmolLM-135M-Instruct) to generate semantically-diverse instances of minority emotion classes; (3) Building balanced data using the integration of the synthetic data with the initial training data; and (4) Fine-tuning and evaluation of RoBERTa-based classifiers through hyperparameter optimization in Optuna by Akiba et al. [1].

The framework proposed particularly focuses on the so-called long-tail response to emotion recognition, in which uncommon classes (e.g. grief, relief) are more frequently poorly recalled. When we produce synthetic samples which are strictly conditioned on these rare labels, then we wish to flatten the curve of the class distribution, yet without any loss of information on majority-classes. Figure 1 represents the entire augmented workflow involving generation and classification whereas Figure 2 represents the typical artificial baseline workflow to perform the comparison.

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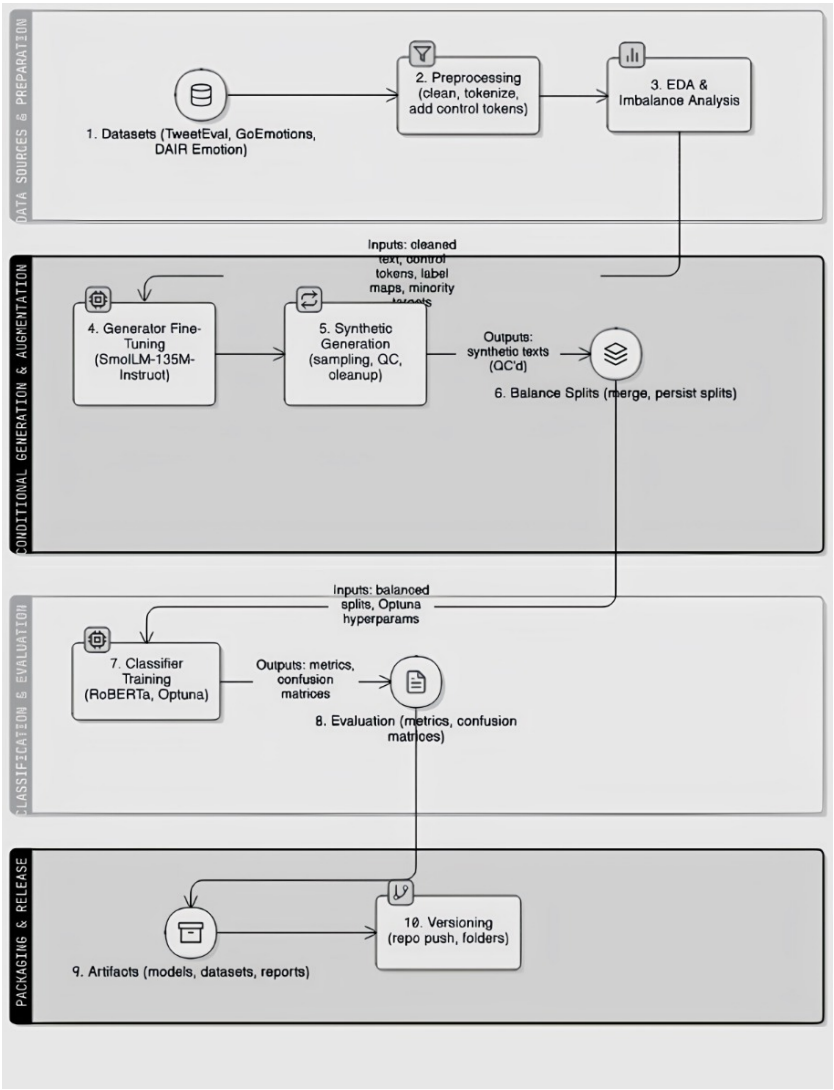

Fig. 1. Pipeline of balanced augmented dataset classification

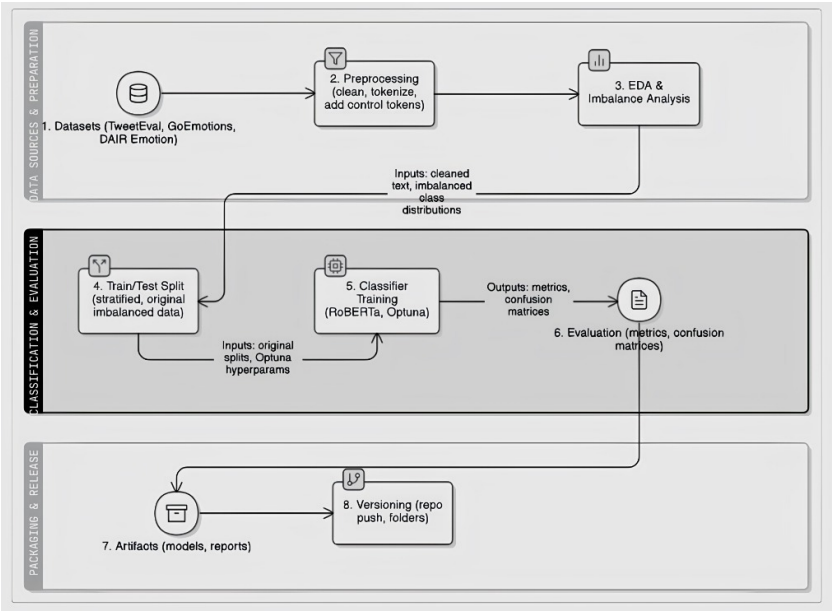

Fig. 2. Pipeline of original non-augmented dataset classification

3.1 Dataset Preprocessing

The pipeline evaluates three benchmarks. GoEmotions by Demszky et al. [5] contains 58,011 Reddit comments annotated with 27 emotions plus neutral and is severely imbalanced, with low-frequency classes such as grief and relief. DAIR Emotion by Saravia et al. [9] includes 20,000 sentences labeled with six emotions. TweetEval by Barbieri et al. [3] is a three-way sentiment classification dataset built from short and noisy social media text.

Preprocessing includes steps of performing lowercasing and deleting of URLs, mentions, hashtags, non-alphanumeric characters and maintaining affective punctuation. Train/validation/test splits are stratified to maintain label distributions and conservative length filtering is used to eliminate trivial or low-information inputs.

3.2 Emotion-Conditioned Text Generation

Minority-class synthesis is performed via emotion-conditioned text generation to reduce label skew. We fine-tune an instruction-tuned generative model (SmolLM-135M-Instruct) using explicit emotion control tokens (e.g., [JOY], [FEAR], [POSITIVE]) to encourage label-faithful generation. Fine-tuning follows a causal language modeling objective with a linear learning-rate schedule, mixed-precision training, and early stopping based on validation performance.

Table 1 details the fine-tuning configuration for SmolLM-135M-Instruct.

Table 1. SmolLM-135M-Instruct Fine-Tuning Configuration

Parameter	Value
Objective	Conditional Causal Language Modeling
Learning Rate	2×10^{-4} (Linear Decay)
Effective Batch Size	64
Epochs	6
Max Sequence Length	128 tokens
Precision	Mixed Precision

3.3 Training Configuration

RoBERTa fine-tuning has constant hyperparameters: the learning rate of 5e -5 (identified with Optuna between 1e -5 and 5e -5), 2 steps of gradient accumulation (effective batch size of 64), warmup ratio of 0.1, weight decay of 0.01, FP16 mixed-precision where available, maximum sequence length of 128 tokens, and AdamW as the optimizer. In the case of TweetEval, Optuna was run on the range of learning rates: 1.495e-05 to 2.030e-05; the highest F1 score of 0.790 was observed with a learning rate of 2.03e-05 with 1 epoch.

Table 2 outlines the training configuration for RoBERTa fine-tuning.

Table 2. RoBERTa Fine-Tuning Training Configuration

Parameter	Value
Learning Rate	5e-5
Gradient Accumulation Steps	2
Effective Batch Size	64
Warmup Ratio	0.1
Weight Decay	0.01
Optimizer	AdamW
Optuna LR Search (TweetEval)	1e-5 to 5e-5
Best Config (TweetEval)	LR: 2.03e-05, Epochs: 1 (F1: 0.790)

3.4 Optimization and Effectiveness

AdamW with a linear learning-rate schedule and weight decay is used for fine-tuning. Hyperparameter optimization is performed with Optuna by Akiba et al. [1] using validation macro-F1 as the objective. Mixed-precision training reduces memory use and improves throughput on constrained hardware provided by Google [6].

3.5 Evaluation Metrics

Assessment is a mixture of both general and task measures. Precision gives an approximate measure yet it is an illusion in case of an imbalance in classes. Macro-F1 gives equal weight to all classes, which ensures that majority classes

do not hide errors of the minority but weighted F1 represents the frequency of classes. Minority-class recall recalls across classes with less than five percent of samples, with a focus on rare emotions.

4 Results and Analysis

4.1 TweetEval Dataset Results

TweetEval by Barbieri et al. [3] is a three-way sentiment classification benchmark on noisy and informal text in social media. RoBERTa base attains the 0.71 accuracy and macro-F1 of 0.71. With the introduction of emotion-conditioned generation to the training set, the performance is enhanced to 0.79 accuracy, as well as 0.79 macro-F1. The strongest improvements are witnessed in the neutral category, which is usually ambiguous, because of heterogeneous contexts. The augmented training data introduces label-faithful neutral samples, which aid the model to decrease confusion on neutral, positive and negative samples.

Table 3 shows the training metrics for TweetEval.

Table 3. SmolLM-135M-Instruct Training Metrics on TweetEval

Model	Training Loss	Eval Loss	Perplexity
SmolLM-135M	3.0846	4.3188	75.10

Table 4 compares the overall performance on the TweetEval dataset.

Table 4. Overall Performance Comparison on TweetEval

Method	Accuracy	Macro F1	Weighted F1
Baseline 1 (No augmentation)	0.72	0.71	0.72
Baseline 2 (Random oversampling)	0.72	0.72	0.72
Ours (Conditional Generation)	**0.79**	**0.79**	**0.79**

Figure 3 displays the confusion matrix for the TweetEval dataset.

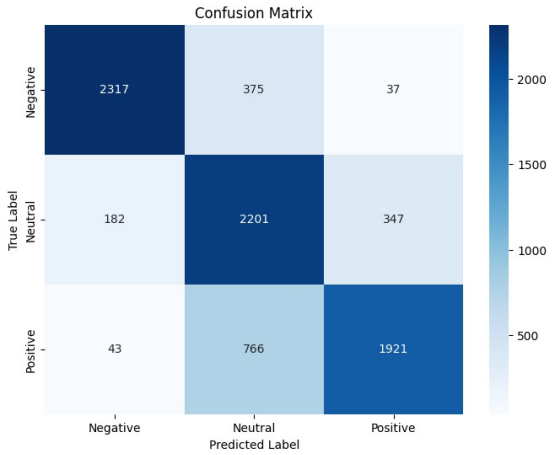

Fig. 3. Confusion matrix for TweetEval (300 DPI)

4.2 DAIR Emotion Dataset Results

DAIR Emotion by Saravia et al. [9] consists of six core emotions. The initial RoBERTa model has 0.93 accuracy and 0.90 macro-F1. As it is augmented, the performance increases to 0.97 and 0.97 accuracy and macro-F1. The gains are more pronounced when it comes to fear and surprise, which are being underrepresented and manifested in a subtle way by the use of linguistic sympathies. The augmentation provides semantically sound and consistent-with-affect samples, which minimize the bias in majority-class.

Table 5 presents the training metrics for the DAIR Emotion dataset.

Table 5. Training and Evaluation Metrics of SmolLM-135M-Instruct on DAIR Emotion

Model	Training Loss	Eval Loss	Perplexity
SmolLM-135M	1.8907	3.9687	52.91

Table 6 summarizes the performance comparison for DAIR Emotion.

Table 6. Overall Performance Comparison on DAIR Emotion Dataset

Method	Accuracy	Macro F1	Weighted F1
Baseline 1 (No Augmentation)	0.93	0.90	0.94
Baseline 2 (Random Oversampling)	0.93	0.91	0.93
Ours (Conditional Generation)	**0.97**	**0.97**	**0.97**

Figure 4 presents the confusion matrix for the DAIR Emotion dataset.

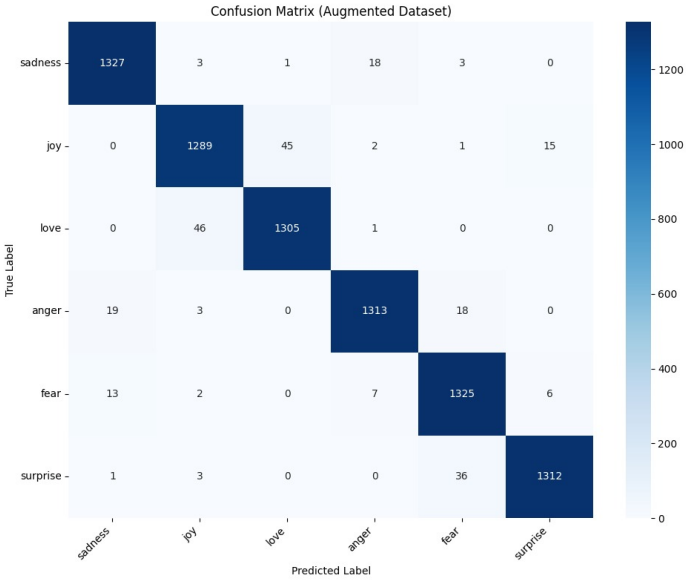

Fig. 4. Confusion matrix for DAIR Emotion (300 DPI)

4.3 GoEmotions Dataset Results

GoEmotions by Demszky et al. [5] is a difficult task because of its fine grained label space and adverse imbalance of the GoEmotions fine grained label space found in Demszky et al. [5]. The baseline model has an accuracy and macro-F1 of 0.58 and 0.51 respectively. As augmentation is done, the performance rises to 0.83 on accuracy and macro-F1. There is a gain observed in both common and unusual labels, such as relief, grief, and admiration. The conditional generator introduces variation in the form of different paraphrases and contexts that seek to maintain the intended feeling, as well as, linguistic diversity.

Table 7 displays the training metrics for the GoEmotions dataset.

Table 7. SmolLM-135M-Instruct GoEmotions Training and Evaluation Metrics

Model	Training Loss	Eval Loss	Perplexity
SmolLM-135M	2.6224	4.1512	63.51

Table 8 provides the overall performance comparison for GoEmotions.

Figure 5 shows the confusion matrix for the GoEmotions dataset.

Table 8. Overall Performance Comparison on GoEmotions

Method	Accuracy	Macro F1	Weighted F1
Baseline 1 (No Augmentation)	0.58	0.51	0.57
Baseline 2 (23-Emotion Model)	0.45	0.46	0.43
Ours (Conditional Generation)	**0.83**	**0.83**	**0.83**

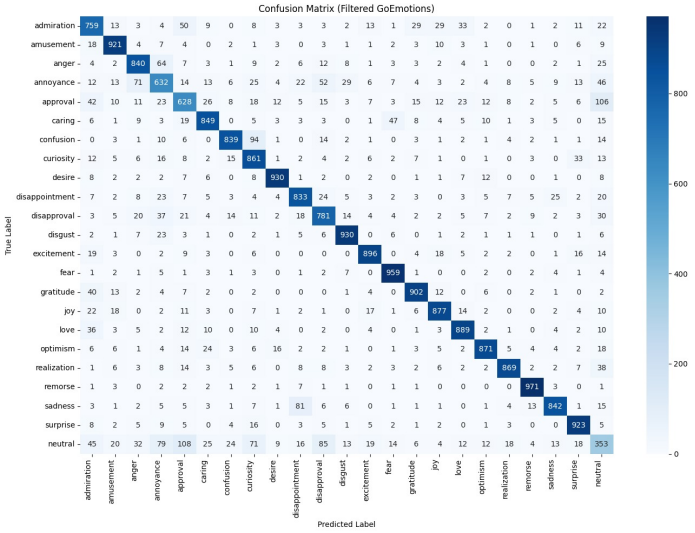

Fig. 5. Confusion matrix for GoEmotions (300 DPI)

4.4 Error Analysis and Confusion Patterns

The confusion matrices and example failure cases are used to analyze error. Despite the dataset improvement, macro-F1 is still weak on semantically neighboring emotions. Confusion is usually involved in GoEmotions between love and joy, between disgust and anger. Fear in DAIR Emotion is occasionally mixed with sadness. These errors are minimized by augmentation especially in the minority classes.

4.5 Performance and Efficiency

Across all three datasets studied by Demszky et al. [5], Barbieri et al. [3], and Saravia et al. [9], there is an improvement in macro-F1 and minority-class recall. The experiments are run with RoBERTa-base with one RTX 4050-GPU with the use of Google Colab (Google [6]). Mixed-precision training can minimize the use of memory and allow bigger effective batch sizes. Collectively with early stopping and tuning with Optuna, training is more efficient compared to untuned baselines and is consistent in regard to performance gains as shown by Akiba et al. [1].

4.6 Limitations

The research has a single GPU resource limit, as well as a limited number of variants of the related models assessed. Although the augmentation performance in rare-category is better, the highly similar emotions are still weak with slight drift, in addition to the very sparse labels are unattainable with limited seed data. Slang, emojis and code-mixing may also influence robustness.

5 Applications, Future Work, and Practices

Interactive systems that are socially responsible are based on the proper recognition of emotions to encourage adaptive conversations. Fine-grained affect is a more realistic phenomenon that can be detected in real time when classifiers are trained on harmonized augmented corpora. The extensions to be developed in the future are multilingual, temporal emotion patterns modeling, and domain adaptation. Large experiments can potentially be achieved through parameter-efficient fine-tuning on resource-constrained devices.

6 Conclusion

To solve the issue of class imbalance and heterogeneity between labels of emotion recognition, we introduce a unified pipeline with emotional text generation that is emotion-conditioned and transformer-based label classification. The approach results in better achievements on GoEmotions, TweetEval, and DAIR Emotion through synthesizing label-faithful samples of sparse emotions and optimizes RoBERTa with Optuna. The closest are the improvements of minority categories that results in a better boundary of decision and generalization.

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References

1. Akiba, T., Sano, S., Yanase, T., Ohta, T., & Koyama, M. (2019). Optuna: A Next-Generation Hyperparameter Optimization Framework. In Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (pp. 2623-2631).
2. Anaby-Tavor, A., Carmeli, B., Goldbraich, E., Kantor, A., Kour, G., Shlomov, S., Tepper, N., & Zwerdling, N. (2020). Do Not Have Enough Data? Deep Learning to the Rescue!. In Proceedings of the AAAI Conference on Artificial Intelligence (Vol. 34, No. 05, pp. 7383-7390).

3. Barbieri, F., Camacho-Collados, J., Espinosa Anke, L., & Neves, L. (2020). TweetEval: Unified Benchmark and Comparative Evaluation for Tweet Classification. In Findings of the Association for Computational Linguistics: EMNLP 2020 (pp. 1644-1650).
4. Chawla, N. V., Bowyer, K. W., Hall, L. O., & Kegelmeyer, W. P. (2002). SMOTE: Synthetic Minority Over-sampling Technique. Journal of Artificial Intelligence Research, 16, 321-357.
5. Demszky, D., Movshovitz-Attias, D., Ko, J., Cowen, A., Nemade, G., & Ravi, S. (2020). GoEmotions: A Dataset of Fine-Grained Emotions. In Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (pp. 4040-4054).
6. Google. (n.d.). Google Colab. https://colab.research.google.com
7. Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., Levy, O., Lewis, M., Zettlemoyer, L., & Stoyanov, V. (2019). RoBERTa: A Robustly Optimized BERT Pretraining Approach. arXiv preprint arXiv:1907.11692.
8. Patrick, A. (2025). Conditional Text Generation and Augmentation for Sentiment Classification (v2). GitHub repository. https://github.com/Adrian-patrick/project-conditional-text-generation-and-augmentation-for-sentiment-classification/tree/v2
9. Saravia, E., Liu, H. C. T., Huang, Y. H., Wu, J., & Chen, Y. S. (2018). CARER: Contextualized Affect Representations for Emotion Recognition. In Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (pp. 3687-3697).
10. Sennrich, R., Haddow, B., & Birch, A. (2016). Improving Neural Machine Translation Models with Monolingual Data. In Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers) (pp. 86-96).
11. Wei, J., & Zou, K. (2019). EDA: Easy Data Augmentation Techniques for Boosting Performance on Text Classification Tasks. In Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP) (pp. 6382-6388).
12. Wolf, T., Debut, L., Sanh, V., Chaumond, J., Delangue, C., Moi, A., Cistac, P., Rault, T., Louf, R., Funtowicz, M., & Brew, J. (2020). Transformers: State-of-the-Art Natural Language Processing. In Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations (pp. 38-45).